

SQL

1. What is SQL?

1. SQL (Structured Query Language) is a standard programming language used to manage and manipulate relational databases. It allows users to create, read, update, and delete data, as well as define database structures and control access to the data.

2. Types of SQL Command

1. ddl : create, drop, truncate, alter, rename
2. dml : insert, update, delete
3. dql : select
4. dcL : grant, revoke
5. tcl : begin, commit, rollback, savepoint, set transaction

3. CRUD Operations

1. Create
2. Read
3. Update
4. Delete

4. SQL Operators

1. Arithmetic Operators
2. Comparison Operators
3. Logical Operators
4. Bitwise Operators

5. Wildcard Characters

1. % (Percent)
2. _ (Underscore)
3. ILike
4. LIKE

6. data types

1. Numeric Data Types

2. Character Data Types
3. Date and Time Data Types
4. Boolean Data Type

7. constraints

1. What is a constraint
2. Types of constraints
 1. NOT NULL Constraint
 2. UNIQUE Constraint
 3. PRIMARY KEY Constraint
 4. FOREIGN KEY Constraint
 5. CHECK Constraint
 6. DEFAULT Constraint
3. Advantages of Constraints
4. Disadvantages of Constraints

8. joins

1. What is a join
2. Types of joins
 1. INNER JOIN
 2. LEFT JOIN
 3. RIGHT JOIN
 4. FULL OUTER JOIN
 5. CROSS JOIN
 6. SELF JOIN
3. Advantages of Joins
4. Disadvantages of Joins

9. subqueries

1. What is a subquery
2. Types of subqueries
 1. Single-row Subquery
 2. Multi-row Subquery
 3. Correlated Subquery
 4. Nested Subquery

3. Advantages of Subqueries

4. Disadvantages of Subqueries

10. clauses

11. What is a clause

12. Types of clauses

1. WHERE Clause

2. GROUP BY Clause

3. HAVING Clause

4. ORDER BY Clause

5. LIMIT Clause

13. Advantages of Clauses

14. Disadvantages of Clauses

15. views

16. What is a view

17. Types of views

1. Simple View

2. Complex View

3. Materialized View

18. Advantages of Views

19. Disadvantages of Views

20. functions

21. What is a function

22. Types of functions

1. Scalar Functions

2. Aggregate Functions

3. User-defined Functions

23. Advantages of Functions

24. Disadvantages of Functions

25. stored procedures

26. What is a stored procedure

27. Advantages of Stored Procedures

28. Disadvantages of Stored Procedures

29. triggers

30. What is a trigger

31. Types of triggers

1. Before Trigger
2. After Trigger
3. Instead of Trigger

32. Advantages of Triggers

33. Disadvantages of Triggers

34. indexes

1. What is an index
2. Types of indexes
 1. Single-column Index
 2. Composite Index
 3. Unique Index
 4. Full-text Index

3. Advantages of Indexes

4. Disadvantages of Indexes

35. CTE (Common Table Expressions)

1. What is a CTE
2. Advantages of CTEs
3. Disadvantages of CTEs

36. Temporary Tables

1. What is a Temporary Table
2. Advantages of Temporary Tables
3. Disadvantages of Temporary Tables

